Team 9

Path 1

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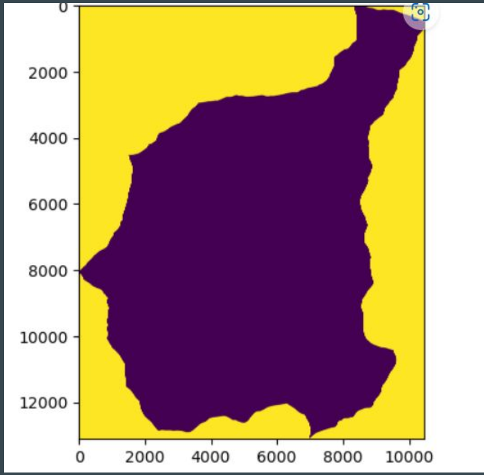
Initial Plan and Thought Process-



The plan was to stack all 3 tif files together into a single tensor.

Using

BareEarth Hillshade (shadow-based terrain visualization) (1 dimension)

BareEarth Dem (elevation data) (1 dimension)

NAIP(ariel images) (4 dimensions)

Stacking these three into a tensor would make a combination of 6 channels of 137 million pixels each. Which would then be split into 256x256 patch size for splitting large dataset into smaller pieces for Pix2Pix input requirements and also for later training use.

Data Preparation

The challenge: Working with the large tensor

- The multiple layers of data were too big into a model

Solution: Breaking Data into Smaller Pieces

- We split the large images into smaller patches so that the model could process them efficiently
- Each patch was 256x256 pixels

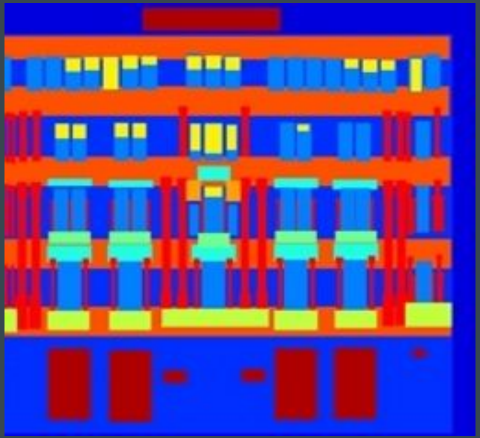
Speaking of pieces:

- Displaying our final road_mask was so demanding on our memory, we had to tile groups of patches to build the final lossless png of the mask.

Pix2pix - What is it

Type of Generative Adversarial Network (GAN) developed in 2017 by UC Berkeley

Had a lot of success for image-to-image translation tasks



Input

Ground truth

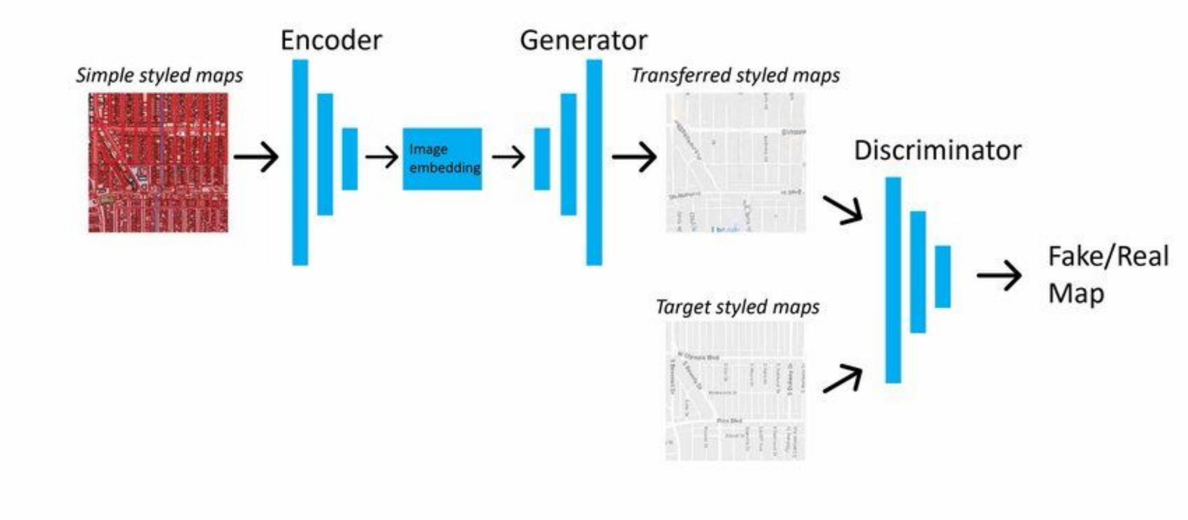
Output

Source: https://github.com/phillipi/pix2pix

Pix2pix - Architecture

Include 2 components:

- Generator: A true artist, takes an input images from one domain and transform it to another domain.
- Discriminator: A judge, evaluates the quality of the image, tell if it is “fake” or “real”.

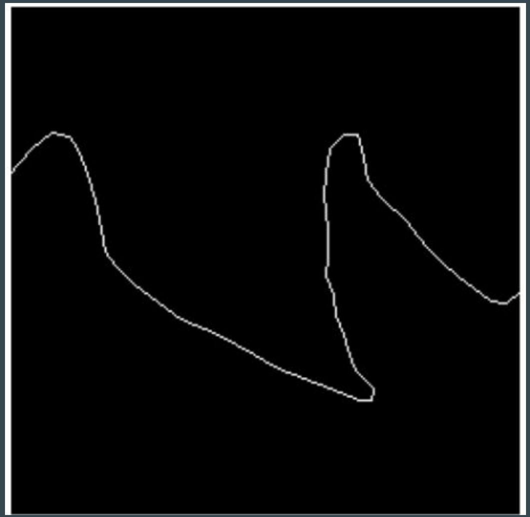


https://arxiv.org/abs/1905.02200

Model Selection - Why Pix2Pix

- We want to frame our problem as translating from the satellite image (1st domain) to the boundary mask (2nd domain).
- Possible to learn the spatial structure of the data and map into boundary mask.

Satellite domain



Boundary domain

Challenges

Challenge 2:

- Memory Usage
- Our dataset was too large to fit in RAM when processing which would cause slowdowns and the kernel to die

Solution:

- Data was stored on a different lab machine with more RAM but no GPU which would mean data needs to transfer to lab with GPU
- In the lab machine with more RAM we pickled the dataset and transferred the pickle files to GPU machine where it would be unpacked and loaded the data in batches for training

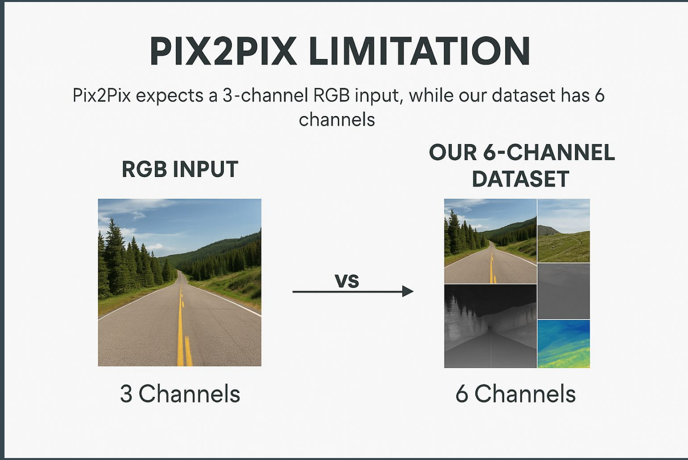
Challenges

Challenge 1:

- Traditional Pix2Pix are intended to transform RGB images to RGB images (dimensions of 3).
- The dataset can be compressed into 6 dimensions output (Hillshade, DEM, R, G, B, Nir)

Our Solutions:

- Modify the model architecture so we can feed in 6-dimension image, and take out 1-dimension mask



Future Improvements

- We could use the RGB channel of NAIP images to feed into the network, and predicting masks on it.
- Exploring other deep learning architectures that naturally support multi-channel inputs like using standalone U-Net without Pix2Pix setup or using DeepLabV3
- Training

My Final Road Mask